

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Fractions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC FRACTIONS · CH2 · L6



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The Map

This lesson collapses Algebraic Fractions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Factorise before cancelling

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use common denominators

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Multiply and divide algebraic fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 State excluded values when needed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Factorise before cancelling

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1H Q12

(a)

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

(b)

$$\frac{2x + 1}{4x} + \frac{7 - 5x}{3x}$$

Finish: (a) $(3y - 4)(2y + 1)$, (b) $\frac{31 - 14x}{12x}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 (a) Factorise fully $6y^2 - 5y - 4$ (2) (b) Express $2\frac{1}{4} \times 7\frac{5}{3} \times x \times x \times x$ as a single fraction in its simplest form. (3)

YOUR SOLUTION

02 Use common denominators

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish: $\frac{1-3x}{x(2x-3)^2}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

24 Express $\frac{4 \cdot 2 \cdot 5 \cdot 3 \cdot 2 \cdot 3 \cdot 9 \cdot 4 \cdot 6 \cdot 17 \cdot 5 \cdot 3 \cdot 2}{x \cdot x \cdot x \cdot x \cdot x}$ divided by $- - +$ as a single fraction in its simplest form.

YOUR SOLUTION

03 Multiply and divide algebraic fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish:

$$\frac{1-3x}{x(2x-3)^2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

24 Express $4\frac{2}{5} - 3\frac{2}{3} + 9\frac{4}{6} - 7\frac{5}{3} - 2x + x + x + x + x$ divided by $- +$ as a single fraction in its simplest form.

YOUR SOLUTION

04 State excluded values when needed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q21

Use trigonometry

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

Factor

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

Finish:

$$\frac{54}{5x + 2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

21 Write $\frac{25}{64} \times \frac{5}{13} \times \frac{6}{8} \times \frac{15}{5} \times \frac{8}{7} \times \frac{2}{2} \times x \times x \times x \times x \times x$ --- times $- + - + - (-)$ as a single fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish:

$$\frac{1-3x}{x(2x-3)^2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

24 Express $\frac{4 \cdot 2 \cdot 5 \cdot 3 \cdot 2 \cdot 3 \cdot 9 \cdot 4 \cdot 6 \cdot 17 \cdot 5 \cdot 3 \cdot 2}{x \cdot x \cdot x \cdot x \cdot x}$ divided by $\frac{-}{- +}$ as a single fraction in its simplest form.

YOUR SOLUTION

06 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q19

Simplify fraction

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

Simplify fraction

$$3x-7 = \frac{21}{4y-3} - 7$$

Finish: $a = \frac{4y-3}{3-2y}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

19 a = 14 3 7 x - x = 7 4 3 y - Express a in the form py q ry s + + where p, q, r and s are integers. Give your answer in its simplest form. a =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Factorise before cancelling
"work out", "show", "hence", a familiar exam phrase	Use common denominators
"work out", "show", "hence", a familiar exam phrase	Multiply and divide algebraic fractions
"work out", "show", "hence", a familiar exam phrase	State excluded values when needed
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Simplify fraction
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.